USER INTERFACE DESIGN EX–3

Difference between CLI (Command Line Interface), GUI (Graphical User Interface) and VUI (Voice User Interface)

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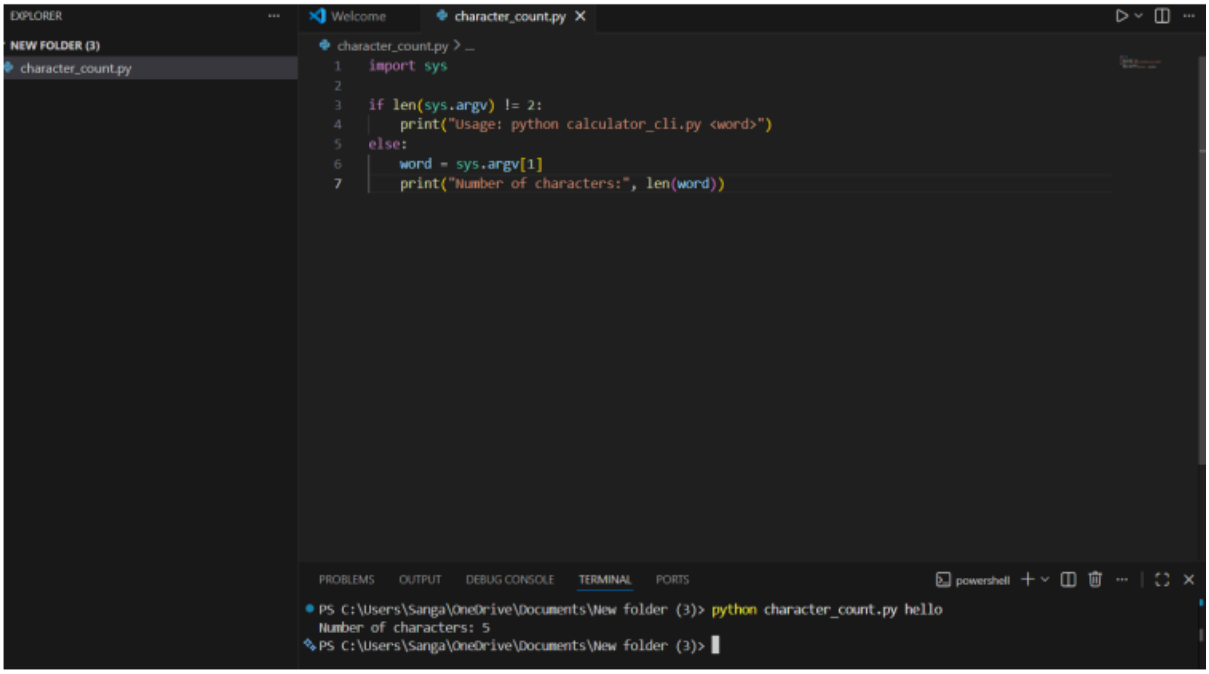
Command Line Interface (CLI)

- A Command Line Interface (CLI) is a text-based method used to interact with a computer system. In this interface, users type commands in a terminal or command prompt to perform operations instead of using graphical elements such as icons or menus.
- CLI is known for being powerful and efficient. Users who are familiar with the command syntax can execute tasks very quickly compared to navigating through multiple graphical menus.
- One major advantage of CLI is its ability to support scripting and automation. By writing scripts, users can automate repetitive operations, which saves time and improves productivity.
- CLI applications are lightweight and require minimal system resources. Because of this, they are commonly used in servers, development environments, and remote system management.
- **Example:** Windows Command Prompt

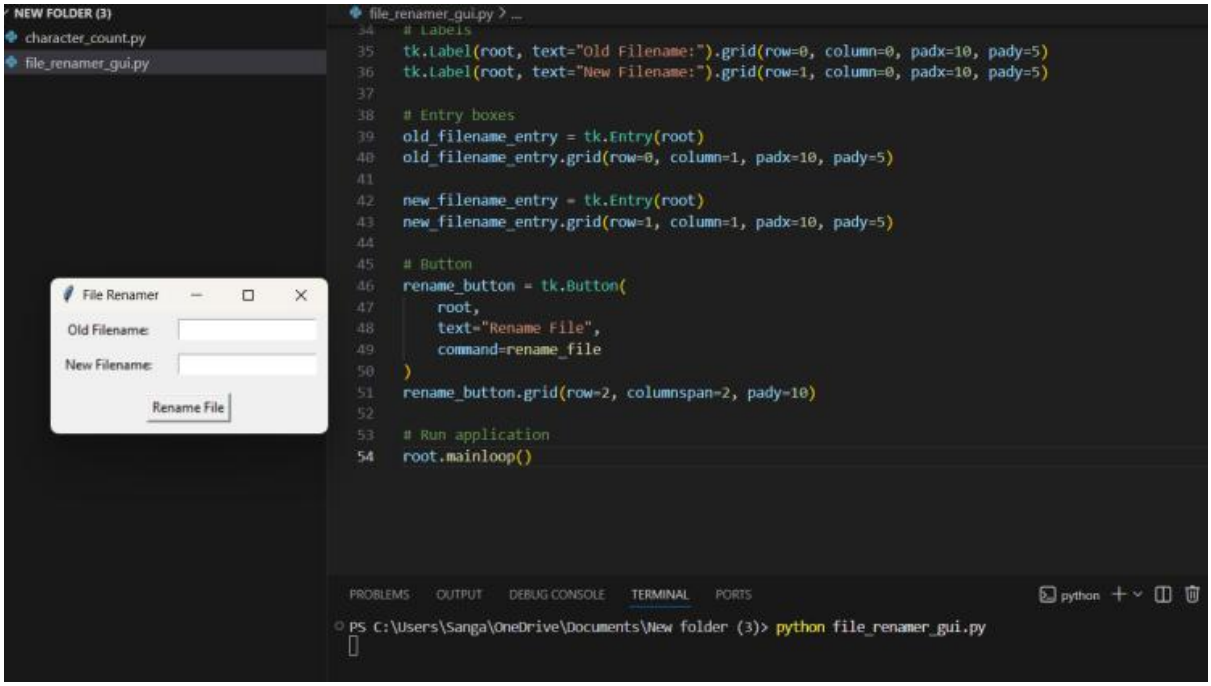
CLI : IMPLEMENTATION AND OUTPUT:



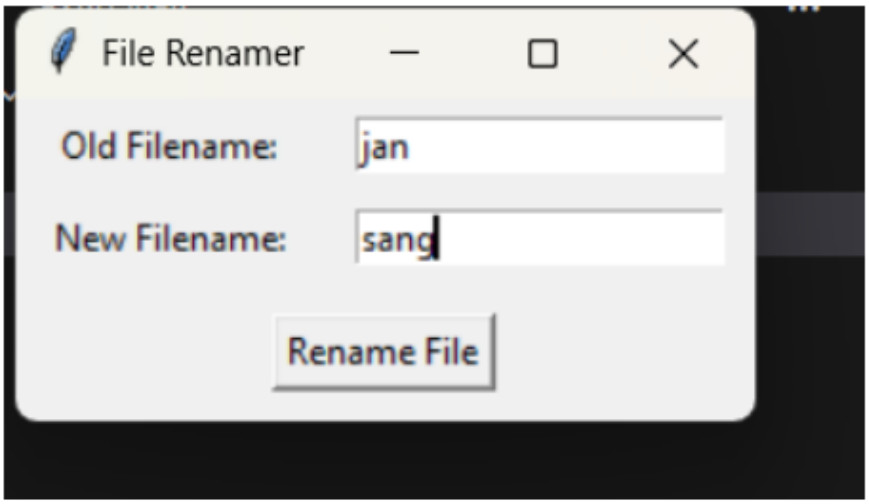
Graphical User Interface (GUI)

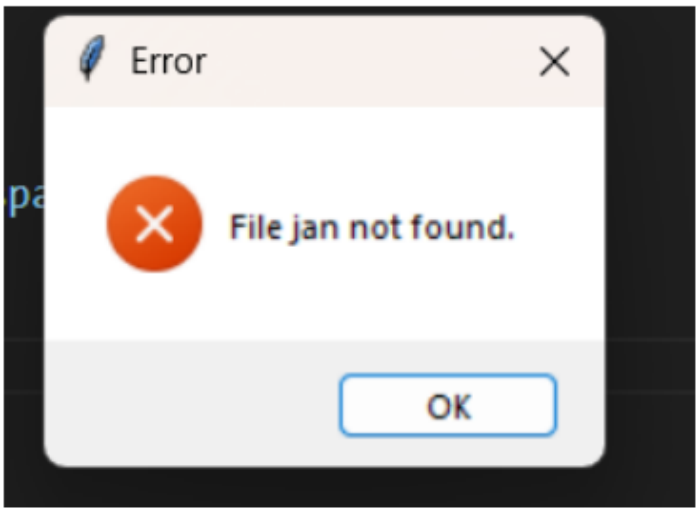
- A Graphical User Interface (GUI) is an interface that allows users to interact with a computer using visual components such as icons, windows, buttons, and menus. It provides a more intuitive and easy-to-understand environment for users.
- GUI systems are designed to be user-friendly, especially for beginners. Instead of memorizing commands, users can perform tasks simply by clicking or selecting graphical options.
- However, GUI applications usually consume more system resources such as memory and processing power. This can sometimes affect performance on low-end devices.
- Another limitation of GUI is that automation is limited when compared to CLI. Many repetitive tasks still require manual interaction, which can make the process slower.
- Example: Windows, Android, iOS

Implementation:



Output:





Voice User Interface (VUI)

- A Voice User Interface (VUI) allows users to interact with computer systems using voice commands. Instead of typing or clicking, users can simply speak to the system to perform tasks.
- One of the major advantages of VUI is hands-free interaction. This makes it very convenient in situations where typing or manual input is difficult.
- VUI also enables more natural communication between humans and machines. The system processes spoken language and converts it into commands that the computer can understand.
- However, voice recognition systems may sometimes face challenges such as background noise, different accents, or unclear speech. These factors can occasionally lead to incorrect command interpretation.
- **Example:** Amazon Alexa, Google Assistant

IMPLEMENTATION:

```python
def listen(prompt):
        text = recognizer.recognize_google(audio)
        print("You said:", text)
        return text
    except:
        print("Could not understand audio")
        return None


# Main program
print("🎤 Welcome to Voice Controlled File Renamer")

old_file = None
while old_file is None:
    old_file = listen("Say the OLD file name (without txt)")

new_file = None
while new_file is None:
    new_file = listen("Say the NEW file name (without txt)")

rename_file(old_file, new_file)
```

OUTPUT:

```
PS C:\Users\Sanga\OneDrive\Documents\uid vui> python voice_file_rename.py
🎤 Welcome to Voice Controlled File Renamer
Say the OLD file name (without txt)
Could not understand audio
Say the OLD file name (without txt)
You said: old
Say the NEW file name (without txt)
Could not understand audio
Say the NEW file name (without txt)
Could not understand audio
Say the NEW file name (without txt)
You said: new new
✅ File renamed from old.txt to new new.txt
```